

Jake Lee

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EDUCATION

Rice University

Bachelor of Science in Computer Science

Houston, TX

Aug 2022 – May 2026

National University of Singapore (NUS)

Study Abroad: Exchange

Singapore

Jan 2025 – May 2025

EXPERIENCE

Celanese Corporation

Software Engineering Intern

May 2025 - August 2025

Houston, TX

- Designed and implemented an enterprise-wide data model using **GraphQL** to record assets, asset properties, and attach real-time data. Led a design review to ensure compliance with technical standards.
- Engineered through **Python** a script to map **1600+** assets to properties, resulting in **8700+** unique combinations in the **Cognite Data Fusion** data model with live field data, resulting in **18% faster querying speeds**.
- Engineered an equipment-health dashboard template to display real-time metrics, historical trends and data patterns, and interactive equipment diagrams. Utilized a **Python Streamlit** frontend and **Cognite Data Fusion** backend with **Cognite SDK** for API calls with **Plotly** and **Pandas** to format and present data.

Rice University Computer Science Department

Teaching Assistant: Computational Thinking

August 2023 - May 2026

Houston, TX

- Assisted students on **Python** programming, search algorithms, basic data structures, mathematical computations. Supported the development of critical thinking skills and problem solving skills in computational thinking.
- Hosted weekly office hours, graded problem sets and exams, answered students' questions, and facilitated academic success. Attended weekly meetings with the professor to discuss course curriculum, planning, and grading.

Plutus Global Inc

Software Engineer

May 2024 - August 2024

San Diego, CA

- Developed a functional prototype of a professional ecommerce website using **React.js** and **HTML/CSS**. Hosted on **Vercel** and implemented with **Tailwind**, **Globe.gl**, **React-Typed** and **Email.js** libraries.
- Worked as a part-time student intern while balancing academic coursework

CyberPatriot

President & Varsity Team Captain

August 2020 - May 2022

San Diego, CA

- Trained program participants for CyberPatriot competitions by creating lessons on cybersecurity topics. Captained the varsity team to 1st place in the CyberPatriot XIV National Competition out of 1000+ teams.
- Designed and created virtual Windows and Linux interfaces on Virtual Machines to simulate cybersecurity vulnerabilities and practice proper solutions and techniques. Held weekly meetings for lessons and practice.

SELECTED PROJECTS

Daily Mood Tracker App | [Github Repository](#) | *Swift, SwiftUI, Firebase, Firestore*

- Designed and engineered a productivity app to track ratings of a user's day, show statistics for user-specific time intervals, and ability to add user-remarks. Consists of 4,000+ lines of code with authentication and data storage.
- Utilized **Swift** and **SwiftUI** for the frontend, **Firebase** and **Firestore** as the backend database, with **Github Releases** to allow distribution (MacOS exclusive).

Terminal Device Driver | *C*

- Engineered a multi-terminal device driver in **C** that allows multiple users to type and receive output across independent terminal windows through communication between the hardware and the operating system.
- Designed a multi-threaded synchronization system following a Mesa-style monitor with deadlock and starvation prevention. Supports I/O operations (read/write) and real-time input reflection and character processing.

OwlDB Messaging App & Document Database | *Golang, HTML/CSS, Typescript, RESTful API*

- Built a web messaging application using **HTML/CSS**, **Typescript** with a **NoSQL**-structured document database with **RESTful API** services using **Golang**, supporting dynamic messaging following a Slack-like format.
- Engineered the backend database to support the ability to create, update, delete, retrieve, and query **JSON** docs. Implemented for concurrent use cases supported by a skip list data structure for efficient insertion and lookup.
- Designed the website to comply with accessibility requirements and used **Typescript** to dynamically update **HTML/CSS** elements to reflect the backend. Included a simple authentication (login/logout) feature.

Memory Allocator | *C*

- Designed and programmed a dynamic memory allocator to replicate **C**'s internal memory structure.
- Used segregated linked lists, a "last in, first out" (LIFO) policy, and list traversal algorithms to optimize memory allocation by maximizing throughput and minimizing space utilization.

TECHNICAL SKILLS

Languages: Python, Java, C, HTML/CSS, JavaScript, TypeScript, Golang, SQL, R, GraphQL, GDScript, Swift

Frameworks: React, JUnit, Tailwind, Hadoop, Apache Spark, Cognite Data Fusion, Streamlit

Libraries: NumPy, Scikit-learn, Pandas, Matplotlib, Ggplot, Plotly

Developer Tools: Git, Github, VS Code, IntelliJ IDEA, RStudio, Azure DevOps, Jupyter Notebook